清华大学本科生考试试题专用纸

考试课程　　　　　　　　　　　　　　　　　年　月　日

Linear Algebra (Int) **Name:** ____________

2024 Fall

Midterm Exam

2024.11.9

Exam Duration: 3 Hours **Student ID:** ____________

This exam includes 6 pages (including this page) and 5 problems. Please check to see if there is any missing page, and then write down your name and student ID number on this page and the first page of your answer sheets. Also write down the initials of your name on the top of every page of your answer sheets, in case they are scattered.

This exam is open book. You are allowed to consult your textbook and notes, and elementary calculators are fine (but nothing with the potential for internet connection is allowed). Plagiarism of all kinds are strictly forbidden and will be severely punished.

Please write down your answers to the problems in the **SEPARATE ANSWER SHEETS**, and follow the following rules:

- **Always explain your answer.** You should always explain your answers. At least write down one sentence to explain your thoughts.
- **Write clearly and legibly.** Make sure that your writings can be read. The graders are NOT responsible to decipher illegible writings.
- **Partial credits will be given.**
- Blank spaces are provided in the exams. Feel free to use them as scratch papers. However, your formal answer has to be written in the **SEPARATE ANSWER SHEETS**, as required by the University.
- The total score of the exam is 36. If your total score exceeds 36 (there are 41 points in total), it will be recorded as 36.

Problem	Points	Score
1	9	
2	10	
3	6	
4	10	
5	6	
Total:	41	

1. Let $A : \mathbb{R}^3 \to \mathbb{R}^3$ be any linear map. We know it fixes all vectors on the plane $x+y+z=0$, i.e., we have $A\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}x\\y\\z\end{bmatrix}$ whenever $x+y+z=0$. And we know A has RREF $\begin{bmatrix}1&2&0\\0&0&1\\0&0&0\end{bmatrix}$. Let's say $A = \begin{bmatrix}\boldsymbol{u} & \boldsymbol{v} & \boldsymbol{w}\end{bmatrix}$.

   (a) (2 points) Write $\boldsymbol{u}-\boldsymbol{v}$ and $\boldsymbol{u}-\boldsymbol{w}$ as A multiplying some vectors, and then calculate the result. (Hint: $\begin{bmatrix}1\\-1\\0\end{bmatrix}$ and $\begin{bmatrix}1\\0\\-1\end{bmatrix}$ are on the plane $x+y+z=0$.)

   (b) (2 points) Consider the RREF of A. What does this say about the relations of $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$?

   (c) (3 points) Find the matrix for A. (Hint: Use the previous subproblems.)

   (d) (1 point) Calculate $A^2 - A$.

   (e) (1 point) Is A invertible? Find this inverse or show why it is not invertible.

2. (This problem has many calculations. A better understanding of matrix multiplications might make most of them a lot faster.) We have three matrices

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

$$B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 2 & 3 \\ 0 & 1 & -2 \end{bmatrix}.$$

$$C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}.$$

(a) (2 points) Which of the following matrix multiplication is legal:

$$ACB, BAC, CAB, CBA.$$

(b) (3 points) Find the inverse of ABC or show that it is not invertible. Find the inverse of BCA or show that it is not invertible.

(c) (4 points) Find the LDU decomposition of $B^{\mathrm{T}}B$, or show that it does not have an LDU decomposition.

(d) (1 point) Calculate $BCAA^{\mathrm{T}}C^TB^{\mathrm{T}} - BCC^TB^{\mathrm{T}}$.

3. We have a matrix $M = \begin{bmatrix} A & AB \\ BA & B \end{bmatrix}$ where A is some 3×2 matrix, and B is some 2×3 matrix.

   (a) (2 points) We can do a block row operation on M to kill the upper right block. Find the matrix for this block row operation, and give me the resulting matrix after this operation.

   (b) (2 points) After the operation in the last subproblem, we can further do a column operation to kill the lower left block. Find the matrix for this block column operation, and give me the resulting matrix after this operation.

   (c) (1 point) Prove that the rank of M is at most 4. In particular, it is not invertible.

   (d) (1 point) Say $BA = I$ but $AB \neq I$. Find the RREF of M. (Write the answer in a block matrix format.)

4. I like linear algebra, LA for short. Let us consider the following matrix, which looks like "LA".

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}.$$

   (a) (3 points) Find its RREF.

   (b) (4 points) Find a basis for $\mathrm{Ran}(A), \mathrm{Ker}(A)$, and give their dimensions.

   (c) (2 points) Find all solutions to $A\boldsymbol{x} = \begin{bmatrix} 1 \\ 3 \\ 2 \\ 3 \\ 4 \end{bmatrix}$. (Please give a complete description of the solution set.)

   (d) (1 point) Find the projection matrix to the kernel of A^{T}. This should be a 5 by 5 matrix.

5. Let V be the solution space to the differential equation $f + 4f'' = 0$. A basis is $\mathcal{B} = (\sin(2x), \cos(2x))$. Another basis is $\mathcal{C} = (\sin(2x + \frac{\pi}{6}), \cos(2x + \frac{\pi}{6}))$.

   Recall that
$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta,$$
$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta.$$

   (a) (2 points) Find the change of coordinate matrices from $\mathcal{B}$ to $\mathcal{C}$ and from $\mathcal{C}$ to $\mathcal{B}$.

   (b) (2 points) Find the coordinates for $\sin(2x + \frac{\pi}{4})$ under basis $\mathcal{B}$ and $\mathcal{C}$.

   (c) (2 points) The derivative map is a map $D : V \to V$, sending inputs $f(x)$ to outputs $f'(x)$. Find the matrix for D where the domain is under basis $\mathcal{C}$, but the codomain is under basis $\mathcal{B}$.